

Mass spectra of elements (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to interpret mass spectra in terms of identity and relative abundance of isotopes.

In 1997, NASA (the National Aeronautics and Space Administration) (and others) launched a mission to study the planet Saturn. The mission, named ‘Cassini’ (<https://solarsystem.nasa.gov/missions/cassini/overview/>), was the first to orbit Saturn with the aim of gaining new insights in the planet, its rings and moons. After spending nearly two decades in space, the mission ended in 2017 when the space probe burned up in Saturn’s atmosphere. The probe carried aboard an Ion and Neutral Mass Spectrometer (INMS) which was able to take samples from the atmosphere of Titan, one of Saturn’s many moons, for the purpose of identifying the elements and compounds in the atmosphere. The atmosphere was found to contain various gases such as nitrogen, methane, hydrogen and argon. In this section, we look at how a mass spectrometer can be used for a less dramatic but equally as important discovery, such as deducing the relative abundances of the isotopes of an element.

In section S1.2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>), you learned how to calculate the relative atomic mass for a sample of an element, but have you wondered how the abundance of isotopes is determined? A mass spectrometer is an instrument that is used to determine the abundance of the various isotopes of an element, based on a measurement of the mass-to-charge (m/z) ratio of ions within a sample of the element. In this section, you will learn how a mass spectrometer is used to determine the relative abundances of the isotopes to calculate the relative atomic mass of an element.

Making connections

Subtopic S3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>) covers how a mass spectrometer can be used to determine the structure of an organic compound.

It should be noted that the operation of a mass spectrometer will not be assessed in the DP chemistry syllabus. It is included here to give an overview of how the relative atomic mass of an element is determined.

A mass spectrometer has five main stages: vaporisation, ionisation, acceleration, deflection and detection.

- Vaporisation: the substance to be analysed is vaporised to produce gaseous molecules.
- Ionisation: high-energy electrons are fired at the gaseous molecules, which causes them to be ionised, forming gaseous ions (known as

molecular ions).

- Acceleration: the gaseous ions are accelerated in an electric field.
- Deflection: the gaseous ions are deflected by an electromagnet. The degree of deflection they undergo depends on their m/z ratio. Ions with lower m/z ratios are deflected the most and ions with higher m/z ratios are deflected the least.
- Detection: the gaseous ions are detected and a mass spectrum is produced.

Figure 1 shows a diagram of a mass spectrometer.

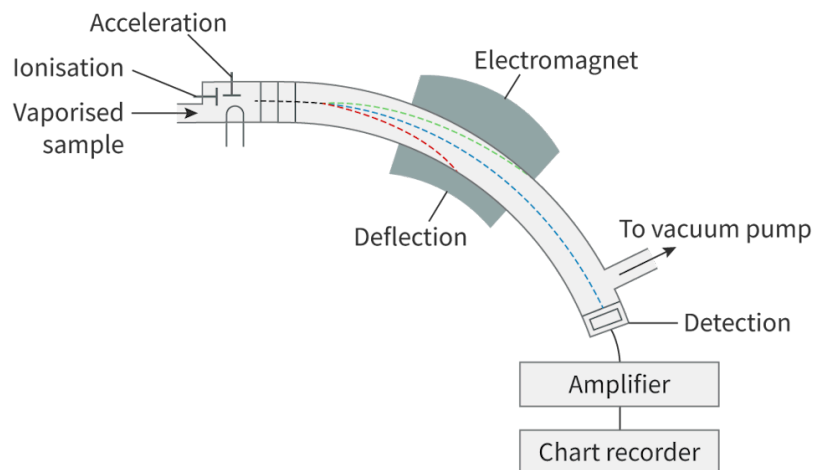


Figure 1. The five stages of a mass spectrometer. Note that the isotope with the lowest m/z ratio is deflected the most (shown in red).

 More information for figure 1

The diagram illustrates the structure and function of a mass spectrometer, with several labeled components. At the left, the diagram begins with a section labeled "Vaporised sample," followed by "Ionisation" and "Acceleration." The structure forms a curved tube leading towards the right. In the central area, there is an "Electromagnet" labeled at the top of the curve, and "Deflection" labeled at the bottom of the curve. There are three dotted lines within the tube: a green line at the top, a blue line in the middle, and a red line at the bottom, all originating from an initial black dotted line. The red dotted line curves towards the "Deflection" area and the green dotted line towards the "Electromagnet" area, while the blue line continues straight. Towards the right end of the tube is an opening labeled "To vacuum pump," and the final section labeled "Detection." Below the detection area are sequentially placed components labeled "Amplifier" and "Chart recorder," indicating the progression of the process.

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The species detected in a mass spectrometer can be individual ions, molecular ions or fragments. Most species detected in a mass spectrometer have a $1+$ charge so the m/z ratio can be used to determine the mass of the species. For example, if the ion $^{204}\text{Pb}^+$ is detected, the m/z ratio is its mass (204) divided by the charge (1) which gives a m/z ratio of 204.

Figure 2 shows the mass spectrum for the element lead (symbol Pb). This mass spectrum shows the percent abundance on the y-axis and m/z ratio on the x-axis.

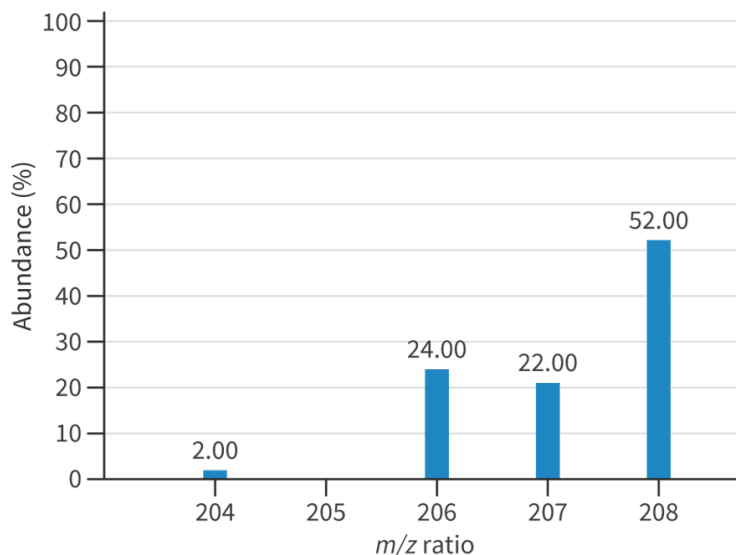


Figure 2. The mass spectrum of lead, Pb.

 More information for figure 2

The image is a graph representing the mass spectrum of lead, with percent abundance on the y-axis and m/z ratio on the x-axis. The y-axis is labeled "Abundance (%)" and ranges from 0 to 100 percent. The x-axis is labeled "m/z ratio" and includes values from 204 to 208. The graph shows four vertical bars corresponding to the isotopes of lead.

The percent abundances for the different m/z values are provided as follows:

- At m/z 204, the abundance is 2%.
- At m/z 206, the abundance is 24%.
- At m/z 207, the abundance is 22%.
- At m/z 208, the abundance is 52%.

This pattern shows a higher abundance of the isotope with m/z 208 compared to the others, indicating a greater presence of this particular isotope in the sample analyzed.

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The data from the mass spectrum is summarised in **Table 1**.

Table 1. Data from mass spectrum of lead, Pb.

Isotope	Abundance (%)
^{204}Pb	2.00
^{206}Pb	24.00
^{207}Pb	22.00

Isotope	Abundance (%)
^{208}Pb	52.00

Before we calculate the relative atomic mass using this data, we will predict its value. The isotopes ^{207}Pb and ^{208}Pb have a combined percent abundance of more than 70% so we can estimate that the relative atomic mass of lead will be closer to 207 or 208 than to 204 or 206.

$$\begin{aligned}\text{Relative atomic mass of lead} &= \frac{(204 \times 2.00) + (206 \times 24.00) + (207 \times 22.00) + (208 \times 52.00)}{100} \\ &= 207.22\end{aligned}$$

So we can see that the answer lies between 207 to 208.

Worked example 1

Figure 3 shows the mass spectrum of neon.

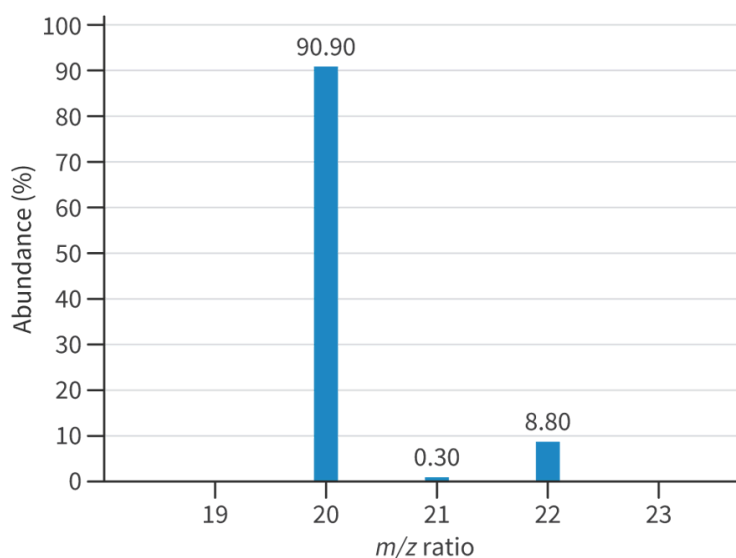


Figure 3. Mass spectrum of neon.

 More information for figure 3

The graph represents the mass spectrum of neon. On the x-axis, it shows the mass-to-charge ratio (m/z) ranging from 19 to 23. The y-axis displays abundance in percentage, ranging from 0 to 100 percent. There are three main data points:

1. At an m/z ratio of 20, the abundance is 90.90%, which is the tallest peak on the graph, indicating the most significant isotope presence.
2. At an m/z ratio of 21, the abundance drops significantly to 0.30%.
3. At an m/z ratio of 22, the abundance is 8.80%, indicating a secondary isotope presence.

The graph visually demonstrates that the predominant isotope of neon has an m/z ratio of 20, followed by a smaller presence of isotopes at m/z ratios of 21 and 22.

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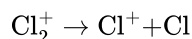
1. Determine the number of naturally occurring isotopes of neon together with their mass numbers and percent abundances.
2. Predict the value of the relative atomic mass of neon based on your answer to part 1.
3. Calculate the relative atomic mass of neon rounded to two decimal places.

1. There are three naturally occurring isotopes with m/z ratios 20, 21 and 22 with percent abundances of 90.90%, 0.30% and 8.80% respectively.
2. The relative atomic mass of neon will be close to 20 as this isotope has the highest percent abundance of 90.90%.
3. Relative atomic mass of neon

$$= \frac{(20 \times 90.90) + (21 \times 0.30) + (22 \times 8.80)}{100} = 20.18$$

Mass spectra of diatomic elements

Now we will consider the mass spectra of diatomic elements such as chlorine, Cl_2 . Elemental chlorine, like all group 17 elements, as well as hydrogen, nitrogen and oxygen, exist as two atoms bonded together to form a diatomic molecule. Chlorine has two naturally occurring isotopes, ^{35}Cl and ^{37}Cl . The ratio of the isotopes is approximately 3:1 in favour of ^{35}Cl . Inside the mass spectrometer the chlorine molecule is ionised producing the molecular ion, Cl_2^+ . This molecular ion can break apart in the mass spectrometer as follows:



The atom of Cl will not be detected in the mass spectrum because it is not charged. The chlorine ion and the molecular ion Cl_2^+ will be detected. **Figure 4** shows the mass spectrum of chlorine gas – before you look, however, predict what the mass spectrum will look like. Note that the mass spectrum shown here has relative intensity on the y-axis rather than percent abundance. Relative intensity is the size of a peak in a mass spectrum relative to the most abundant ion which is shown as the tallest peak in the spectrum. The most abundant ion is given an intensity of 100 and is known as the base peak.

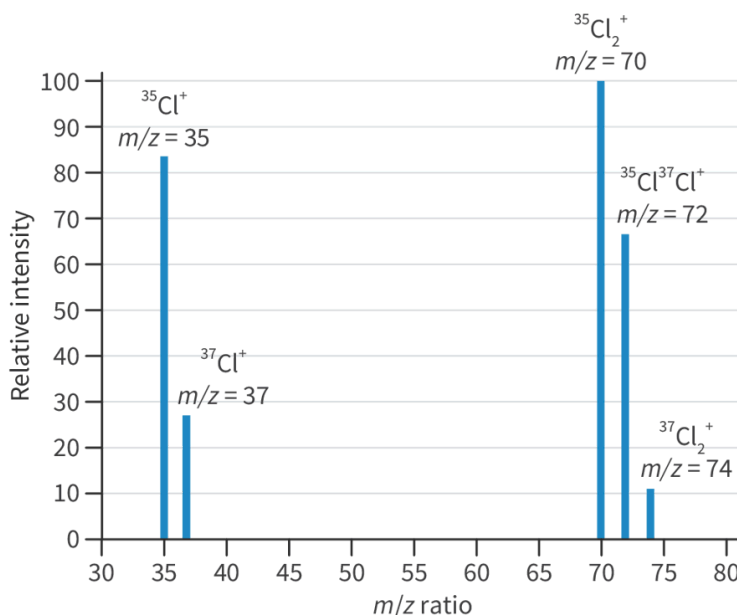


Figure 4. The mass spectrum of the diatomic element chlorine, Cl_2 .

 More information for figure 4

The image is a mass spectrum graph of chlorine with the x-axis labeled as "m/z ratio" and ranging from 30 to 80, while the y-axis is labeled as "Relative intensity" with values from 0 to 100. There are several vertical bars at different points along the x-axis indicating the relative intensity of various isotopes.

1. A bar at m/z 35 with the highest relative intensity of 77 is labeled as " $^{35}\text{Cl}^+$ ".
2. Another bar at m/z 37 with a lower intensity of 25 is labeled as " $^{37}\text{Cl}^+$ ".
3. At m/z 70, a bar with an intensity of 57 is labeled as " $^{35}\text{Cl}_2^+$ ".
4. At m/z 72, another bar with an intensity of 38 is labeled as " $^{35}\text{Cl}^{37}\text{Cl}^+$ ".
5. Lastly, at m/z 74, a bar with the lowest intensity of 8 is labeled as " $^{37}\text{Cl}_2^+$ ".

The graph illustrates the isotopic composition of chlorine and the distribution of its ions in the mass spectrum.

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Looking at this mass spectrum we see more peaks than expected. This occurs because some chlorine molecules consist of two atoms of the isotope ^{35}Cl , some molecules consist of two atoms of the isotope ^{37}Cl and some a combination of both isotopes (one atom of ^{35}Cl and one atom of ^{37}Cl). Next, we will look at each individual peak and the ions responsible for their production.

The peaks at m/z ratios 35 and 37 are caused by the individual chlorine ions $^{35}\text{Cl}^+$ and $^{37}\text{Cl}^+$. They are detected as individual ions because the molecular ion Cl_2^+ breaks apart in the mass spectrometer. Note the height of the two peaks are 3:1 according to the relative abundance of the two isotopes ^{35}Cl and ^{37}Cl .

The peak at m/z 70 is caused by the detection of a molecular ion composed of two ^{35}Cl atoms $^{35}\text{Cl}_2^+$. The peak at m/z 72 is caused by the detection of the molecular ion composed of one atom of ^{35}Cl and one atom of ^{37}Cl , $^{35}\text{Cl}^{37}\text{Cl}^+$. Finally, the peak at m/z 74 is caused by the detection of the molecular ion composed of two atoms of ^{37}Cl , $^{37}\text{Cl}_2^+$.

In this activity you will interpret the mass spectra of some elements.

Activity

- **Learner profile attribute:** Inquirer
- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual

Part 1

Use **Figure 5** to answer the following questions:

1. State the number of isotopes in the natural occurring sample of the element.
2. Predict the relative atomic mass of the element based on the percent abundances of the isotope.

3. Summarise the data from the mass spectrum in a table.
4. Calculate the relative mass of the element to two decimal places. Compare the value to your prediction in question 2.

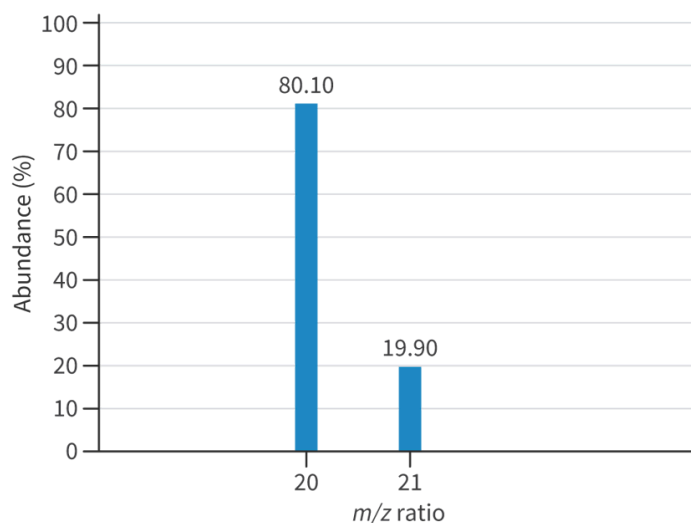


Figure 5. Mass spectrum of an element.

© More information for figure 5

Part 2

Sketch a diagram of the mass spectrum of zirconium given the following data:

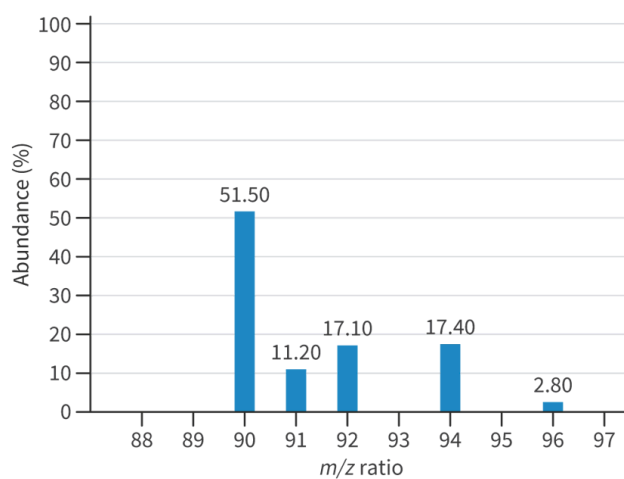
Isotope	
^{90}Zr	
^{91}Zr	
^{92}Zr	
^{94}Zr	
^{96}Zr	

Part 1

1. Two naturally occurring isotopes.
2. The relative atomic mass will be close to 20.
3. See **Table 2** below.
4. $A_r = 20.20$

Table 2. Data from mass spec

Isotope	
^{20}X	
^{21}X	

Part 2

Mass spectrum of zirconium.

